Demonstration Order in In-Context Learning: Real, Task-Dependent, and Often Mismeasured

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Abstract

Few-shot prompting is the default way to adapt large language models, yet the order of the demonstrations is rarely controlled. We ask how demonstration order affects in-context learning as a function of shot count, task type and model size, and which positions drive the output. We run one controlled protocol on five QWEN2.5-INSTRUCT models (0.5B–14B) across sentiment (SST-2), natural language inference (RTE, MNLI) and dialogue summarization (SAMSUM), scoring about 2.36M prompts with deterministic decoding. We measure order sensitivity with a pairwise gap whose expectation does not depend on the number of sampled orders, estimate per-position influence with causal interventions, and check whether order effects survive split-half reliability tests and an identical-prompt numerical noise floor. Order variance is already at its plateau at two shots; the familiar "rise until four shots" shape of the best–worst range is an artifact of how many orders are available. NLI is consistently more order-sensitive than sentiment (mean best–worst range 5.9–13.3 points on MNLI vs. 1.0–4.8 on SST-2 at $K \geq 4$). Position effects are not a universal recency/primacy split: "last demo $\approx$2–3$\times$ first" holds in only 2 of 15 model–task cells, mid-size models show anti-recency on NLI, and summarization shows recency for small models rather than primacy. Scale reduces order sensitivity on SST-2 and RTE (Spearman $\rho = -1$), but not on MNLI or SAMSUM. For strong models on easy tasks, "best orders" cannot be told apart from item-level noise, so order search there overfits; on harder tasks, order effects are systematic and order-averaging is justified.

1 Introduction

Give a language model the same four demonstrations in a different order, and it may give a different answer. Few-shot prompting Brown et al. [2020] is how most practitioners adapt large language models (LLMs) to a new task, but the order of the demonstrations is almost never treated as a variable. It is fixed by whatever order the examples happened to be listed in. If order swings accuracy by many points, then reported few-shot numbers carry hidden variance, and practitioners need to know when order search, calibration or order-averaging is worth the effort.

Prior work shows that order can matter a great deal. Lu et al. [2022] found that some permutations of the same demonstrations give near state-of-the-art accuracy with GPT-2/GPT-3 while others are close to chance. Zhao et al. [2021] documented a recency bias: GPT-3 tends to repeat the label of the last demonstration. Liu et al. [2024] found U-shaped position effects in long contexts. From these results a common picture has formed: order sensitivity grows with the number of shots up to a point, classification favors the most recent demonstration, generation favors the first, NLI is much more fragile than sentiment, and larger models are more robust.

Does this picture hold for modern instruction-tuned models, and is it measured correctly? We see three gaps. First, most studies report the best–worst range over the sampled orders. The range grows with the number of draws, and the number of possible orders ($K!$) grows with the number of shots K, so trends in the range over K mix model behavior with sampling. Second, position

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influence is usually inferred from observational correlations (e.g., how often the prediction matches the last label), which are confounded with demonstration content. Third, few studies ask whether the "best" order is a real property of the prompt or a lucky draw on a small test set, and none we know of separate order effects from the numerical non-determinism of batched bf16 inference.

We address these gaps with a single controlled protocol on the QWEN2.5-INSTRUCT family (0.5B, 1.5B, 3B, 7B, 14B) across sentiment (SST-2), two NLI tasks (RTE, MNLI) and dialogue summarization (SAMSUM). We test five specific hypotheses (H1–H5, section 3). We measure sensitivity with an expected pairwise gap (PAIRGAP) that does not depend on the number of sampled orders. We estimate per-position influence *causally*, by flipping the label of one demonstration or inserting a style marker into one demonstration's summary and measuring the change in the model's output distribution. We test whether order effects reproduce across halves of the test set, and we re-run identical prompts to measure the numerical noise floor.

Most of the specific hypothesis does not survive. Order variance is at its plateau already at $K = 2$; the "rise until 4-shot" shape appears only in the range, and tracks the number of available orders (2, 6, then 8). NLI is more order-sensitive than sentiment for every model, but the thresholds fail: the mean best–worst range at $K \geq 4$ is 5.9–13.3 points on MNLI (not > 15). "Last demonstration $\approx$2–3$\times$ the first" holds in only 2 of 15 model–task cells; 1.5B and 3B models show *anti-recency* on NLI, and summarization shows *recency* (first/last influence ratio 0.64–0.80) rather than primacy. Larger models are less order-sensitive on SST-2 and RTE (PAIRGAP falls from 2.10 to 0.39 points on SST-2), but not on MNLI, and on SAMSUM the 14B model is the most order-sensitive. Finally, for models of 3B and above on SST-2, the best and worst orders cannot be told apart from re-shuffled noise (observed/null range ratio 0.89–1.00).

Our contributions are:

- We conduct a controlled study of demonstration order across 6 shot counts, 4 tasks of 3 types and 5 model sizes from one family, with about 2.36M scored prompts and deterministic decoding.
- We show that the commonly reported best–worst range produces a spurious "rise then plateau" trend over shot count, and propose using an expected pairwise gap that is comparable across K.
- We measure per-position influence with causal label-flip and style-marker interventions, and find that the direction of position effects depends on both task and model size, with no universal recency/primacy split.
- We complement sensitivity numbers with split-half reliability tests and an identical-prompt noise floor, which show when a "best order" is real and when choosing one would overfit.

Section 2 reviews related work, section 3 describes our protocol, section 4 reports results for each hypothesis, and section 5 discusses implications and limitations.

2 Related Work

Order sensitivity in in-context learning. In-context learning with demonstrations was popularized by GPT-3 Brown et al. [2020]. Lu et al. [2022] showed that permuting the same demonstrations can move GPT-2 and GPT-3 classification accuracy from near chance to near state of the art, that this sensitivity did not shrink with model size, and that good orders do not transfer across models; they proposed selecting orders with entropy statistics on a synthetic probing set. Zhao et al. [2021] traced part of this instability to majority-label, recency and common-token biases, and corrected them with contextual calibration. Unlike these studies, which used base models of the GPT-2/3 era and mostly reported ranges over sampled permutations, we study a modern instruction-tuned family, use a sensitivity metric that is unbiased in the number of sampled orders, and test whether order effects exceed item-level and numerical noise.

Position effects in long contexts. Liu et al. [2024] found that LLMs use information at the start and end of long contexts better than information in the middle. That work studies retrieval of one relevant passage among many distractors. We instead study short prompts in which every demonstration is relevant, and ask how much each position steers the output. We measure this with direct interventions on one position at a time rather than by moving a single relevant item.

What demonstrations contribute. Min et al. [2022] found that the format and label space of the demonstrations matter more than whether their labels are correct. Our label-flip probe builds on this observation: it asks whether flipping a single label, at a given position, moves the model toward that

label. We find that it often does, but that for mid-size models on NLI a flipped final label moves predictions *away* from it. Work on demonstration *selection* Liu et al. [2022], Rubin et al. [2022] chooses which examples to show; we hold the selected set fixed and vary only the order, so our results describe the variance that remains after selection.

Evaluation variance. Prompt-format choices can change LLM accuracy by large margins Sclar et al. [2024], which suggests that many reported gains are within the noise of prompt design. We contribute two checks for the order dimension of this problem: a split-half reliability test that asks whether a good order stays good on held-out test items, and an identical-prompt rerun that measures the non-determinism of batched bf16 inference in vLLM Kwon et al. [2023].

3 Method

3.1 Problem formulation and hypotheses

A K-shot prompt consists of an instruction, a demonstration set $S = \{d_1, \ldots, d_K\}$ drawn from a training pool, an order π over S, and a test input x. For a test set $\mathcal{D}_{\text{test}}$ and a metric m (accuracy or ROUGE-L), let $m(S, \pi)$ be the score of the model on $\mathcal{D}_{\text{test}}$ when every test prompt uses demonstrations S in order π. Because we decode deterministically, any difference between $m(S, \pi)$ and $m(S, \pi')$ is caused by order (up to numerical noise, which we measure in section 3.2).

We test five hypotheses:

H1 Order variance increases with K up to $K = 4$, then plateaus.

H2 For classification, the last demonstration is 2–3$\times$ as influential as the first (recency).

H3 For generation, the first demonstration is more influential than the last (primacy).

H4 The best–worst order gap exceeds 15 accuracy points on NLI and stays below 5 on sentiment.

H5 Larger models are less order-sensitive.

3.2 Measuring order sensitivity

For each demonstration set S we evaluate a set Π_S of distinct orders and compute three statistics, which we then average over demonstration sets: the standard deviation of $m(S, \pi)$ over $\pi \in \Pi_S$; the range $\max_\pi m(S, \pi) - \min_\pi m(S, \pi)$; and the expected pairwise gap

$$\textsc{PairGap}(S) = \frac{1}{|\Pi_S|(|\Pi_S| - 1)} \sum_{\pi \neq \pi' \in \Pi_S} \big|m(S, \pi) - m(S, \pi')\big|. \tag{1}$$

PairGap estimates $\mathbb{E}|m(S, \pi) - m(S, \pi')|$ for two random orders. Its expectation does not depend on $|\Pi_S|$, so it can be compared across K even though $K = 2$ has only two orders and $K = 3$ has six. The range, in contrast, is the maximum minus the minimum of $|\Pi_S|$ draws, and grows with $|\Pi_S|$ even when the underlying distribution is fixed. For classification we also report the *flip rate*: the fraction of test items whose prediction changes between two orders.

Is the best order real? On a test set of 200 items, two orders can differ by a few points by chance. We therefore compute a *split-half reliability*: we split $\mathcal{D}_{\text{test}}$ into two random halves, score every order on each half, and compute the Spearman correlation of the per-order scores across halves (averaged over 20 splits). We compare the observed range to a *null range* obtained by shuffling order labels independently within each test item. The null keeps item-level variability and removes any order-level structure. A ratio near 1 means the best and worst orders are indistinguishable from noise.

Numerical noise floor. Batched bf16 inference in vLLM is not bit-deterministic across batch compositions. We re-run the exact same prompts ($K = 4$, demonstration sets 0–2, all 8 orders) and define the *noise share* as $\mathbb{E}[(\text{rerun difference})^2]/\mathbb{E}[(\text{order difference})^2]$, the fraction of order-induced squared score differences that identical reruns also produce.

3.3 Measuring position influence causally

Observational estimates (e.g., regressing the prediction on the label at each position) are confounded with demonstration content when only 10–20 demonstration sets are available. We therefore intervene on one position at a time, keeping everything else fixed (algorithm 1).

Algorithm 1 Causal position-influence probe

Input: demonstration sets $\{S_j\}$, orders $\{\pi_{j,1}, \pi_{j,2}\}$, test set $\mathcal{D}_{\text{test}}$, intervention ϕ
for each set S_j and order $\pi \in \{\pi_{j,1}, \pi_{j,2}\}$ **do**
  $p_0(x) \leftarrow \log P(\text{target} \mid S_j, \pi, x)$ for all $x \in \mathcal{D}_{\text{test}}$ ▷ *unmodified prompt*
  for position $i = 1, \ldots, K$ **do**
    $S_j^{(i)} \leftarrow \phi(S_j, \pi, i)$ ▷ *flip label / add marker at position i only*
    $\Delta_{j,\pi,i} \leftarrow \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_x \left[\log P(\text{target} \mid S_j^{(i)}, \pi, x) - p_0(x) \right]$
return influence$_i$ = mean of $\Delta_{j,\pi,i}$ over (j, π), with bootstrap CI over j

Table 1: Tasks, demonstration pools (length-filtered training sets) and test sets.

Task	Type	Demo pool	Test items
SST-2	sentiment, 2-way	41,584	200 (val, balanced)
RTE	NLI, 2-way	1,830	200 (val)
MNLI (matched)	NLI, 3-way	327,814	201 (val, balanced)
SAMSUM	dialogue summarization	7,640	80 (test)

Label-flip probe (classification). At $K \in \{4, 8\}$ we flip the label of the demonstration at position i to a different label y' (chosen with a fixed seed) and define the influence of position i as the mean increase in $\log P(y' \mid \text{prompt})$ over test items. A recency effect predicts that influence grows with i; H2 predicts a last/first ratio of 2–3.

Style-marker probe (generation). At $K \in \{4, 6\}$ we rewrite the summary of demonstration i to begin with "In short, " and define influence as the increase in $\log P(\text{output begins with "In short,"})$ relative to a control prompt with no marker. H3 predicts that influence is largest at position 1.

Both probes use demonstration sets 0–9 and the first two orders of each. We report 95% confidence intervals from 2,000 bootstrap resamples of demonstration sets.

3.4 Experimental setup

Models and decoding. We use QWEN2.5-INSTRUCT at 0.5B, 1.5B, 3B, 7B and 14B parameters Qwen Team [2024] in bf16, served locally with vLLM Kwon et al. [2023] on one shared NVIDIA RTX A6000. Using one family keeps model size as the only systematic difference. For classification, the next token is restricted to the label tokens, giving a log-softmax over the label set; the prediction is the argmax. For generation we use greedy decoding with at most 80 new tokens, stopping at a newline.

Prompts. We use plain-text completion prompts without a chat template, with one template per task: SST-2 uses `Review: ... / Sentiment: positive|negative`; RTE uses `Premise / Hypothesis / Answer: yes|no`; MNLI uses `yes|maybe|no`, with an instruction defining them as entailment, neutral and contradiction; SAMSUM uses `Dialogue: ... / Summary: ...` under the instruction "Summarize each dialogue in one or two sentences."

Data. Table 1 lists the tasks. We use SST-2 Socher et al. [2013], RTE and MNLI from GLUE Wang et al. [2018], Williams et al. [2018], and SAMSUM Gliwa et al. [2019].

Design. For classification, $K \in \{1, 2, 3, 4, 6, 8\}$. For each K we draw 10 random demonstration sets (labels not forced to balance, so label positions vary) and evaluate each under $\min(K!, 8)$ distinct orders: all orders for $K \leq 3$ and 8 random orders for $K \geq 4$. For the 0.5B, 1.5B and 3B models we add 10 extension sets per K (seed 1042); we skip these for 7B and 14B to save GPU time on the shared host. For SAMSUM, $K \in \{1, 2, 3, 4, 6\}$, 6 sets, $\min(K!, 6)$ orders and 80 test dialogues. The 14B probes are run only at $K = 8$ (classification) and $K = 6$ (SAMSUM). Zero-shot ($K = 0$) and the majority class serve as baselines. In total we score about 2.36M prompts and generations.

Metrics and statistics. We report accuracy for classification and stemmed ROUGE-L F1 Lin [2004] for SAMSUM. We test trends over K and over model size with Spearman correlations and report 95% bootstrap confidence intervals over demonstration sets.

Table 2: Accuracy (%) and ROUGE-L by shot count K, pooled over all demonstration sets and orders. $K = 0$ is zero-shot. Majority-class accuracy is about 50 (SST-2, RTE) and 33 (MNLI). Best per column in **bold**.

	SST-2 (acc)				RTE (acc)				MNLI (acc)				SAMSUM (ROUGE-L)			
Model	K=0	1	4	8	0	1	4	8	0	1	4	8	0	1	4	6
0.5B	89.5	86.8	90.1	90.8	60.5	57.6	69.0	69.6	48.0	46.2	54.7	57.0	19.9	24.8	20.9	21.8
1.5B	92.5	93.5	93.5	93.9	77.5	77.8	82.6	83.1	64.0	64.6	65.5	66.1	27.3	32.0	34.8	34.9
3B	90.0	92.4	93.2	93.3	84.5	82.6	84.9	85.0	74.0	78.9	81.3	81.4	27.0	32.2	33.7	34.1
7B	91.5	**93.6**	93.3	93.4	88.0	84.5	86.1	85.7	**81.0**	**83.3**	**83.3**	**84.0**	**33.5**	36.0	36.0	37.5
14B	**94.0**	**93.6**	**94.4**	**94.4**	**88.5**	**89.6**	**91.1**	**91.2**	80.0	79.0	77.8	82.5	31.9	**37.6**	**38.4**	**38.6**

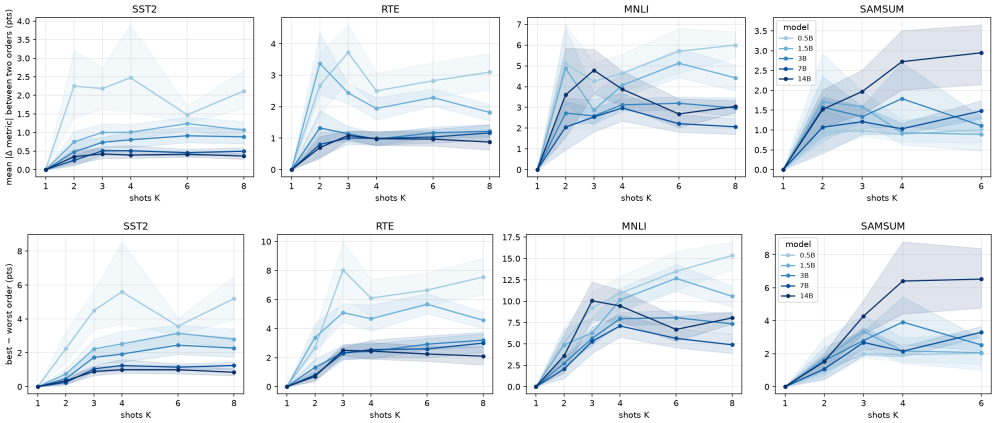


Figure 1: Order sensitivity vs. shot count K for each task and model (shaded: 95% bootstrap CI over demonstration sets). *(Top)*Mean pairwise gap (PAIRGAP), whose expectation does not depend on the number of orders sampled: from $K = 2$ the curves are roughly flat. *(Bottom)*Best–worst range over the sampled orders: it shows a "rise until $K = 3$–4, then plateau" shape, which tracks the number of available orders (2 at $K = 2$, 6 at $K = 3$, 8 for $K \geq 4$) rather than the model.

4 Results

Demonstrations help small models most. Table 2 gives the accuracy context for all later results. All models are well above the majority class. Demonstrations add the most for small models (0.5B on RTE: 60.5 zero-shot vs. 69.6 at $K = 8$) and little for large ones (14B on SST-2: 94.0 vs. 94.4). On MNLI, 14B accuracy *drops* below zero-shot at $K = 4$ (77.8 vs. 80.0) and recovers at $K = 8$ (82.5), so a few unbalanced demonstrations can hurt a strong model.

4.1 H1: order variance is flat from two shots on

Does order variance rise until 4 shots and then plateau? Table 3 and figure 1 (top) show that it does not. Measured with PAIRGAP, order variance is already at its plateau at $K = 2$. Over $K = 2$–4 there is a weak upward trend on SST-2 and MNLI ($\rho \approx 0.16$–0.17, $p < 0.02$), but the K-means differ by less than 0.5 points; RTE and SAMSUM show no trend. Per model, only 1 of 20 task–model curves rises significantly over $K = 2$–4 (SST-2, 3B). For $K \geq 4$, 3 of 20 curves have a significant trend, with mixed signs (MNLI 0.5B up, MNLI 7B down, SAMSUM 7B up).

The best–worst range, in contrast, shows exactly the hypothesized shape (figure 1, bottom): on MNLI, for instance, the 0.5B range climbs from about 5 points at $K = 2$ to about 11 at $K = 4$. This shape follows the number of orders we can sample (2, then 6, then 8), because the maximum minus the minimum of more draws is larger. **H1 is not supported**; the apparent support is a measurement artifact. Studies that compare ranges over all $K!$ orders, or over a fixed small number of permutations, across shot counts should be read with this in mind.

Table 3: H1: mean pairwise order gap (PAIRGAP, points) by shot count, pooled over models, with Spearman correlations between K and the gap for $K \leq 4$ and $K \geq 4$. $K = 1$ has a single order and zero variance by construction. From $K = 2$ the gap is essentially flat.

Task	K=2	3	4	6	8	ρ ($K \leq 4$) [p]	ρ ($K \geq 4$) [p]
SST-2	0.94	1.10	1.19	1.02	1.12	0.17 [0.007]	0.03 [0.63]
RTE	2.03	2.09	1.60	1.82	1.79	−0.02 [0.71]	0.07 [0.26]
MNLI	3.88	3.34	3.81	4.12	3.99	0.16 [0.016]	−0.02 [0.78]
SAMSUM (ROUGE-L)	1.36	1.42	1.47	1.49	–	0.09 [0.40]	0.07 [0.61]

Table 4: H4: order sensitivity at $K \geq 4$ (8 orders per demonstration set; 6 for SAMSUM). "Mean range" averages best–worst over demonstration sets; "max range" is the worst single set. Units are accuracy points (ROUGE-L points for SAMSUM). Lowest (least order-sensitive) per row in **bold**. Only the worst MNLI sets for 0.5B and 1.5B exceed 15 points.

Task	Statistic	0.5B	1.5B	3B	7B	14B
MNLI	mean range	13.3	11.1	7.8	**5.9**	8.1
	max range	21.7	17.5	11.7	**9.8**	11.7
	SD	4.7	4.0	2.7	**2.1**	2.8
	PAIRGAP	5.5	4.5	3.1	**2.4**	3.2
RTE	mean range	6.8	5.0	2.9	2.7	**2.3**
	max range	12.5	8.5	5.2	5.2	**4.0**
	SD	2.4	1.7	1.0	0.9	**0.8**
	PAIRGAP	2.8	2.0	1.1	1.1	**0.9**
SST-2	mean range	4.8	2.8	2.2	1.2	**1.0**
	max range	15.5	6.7	6.0	2.0	**1.5**
	SD	1.8	1.0	0.8	**0.4**	**0.4**
	PAIRGAP	2.0	1.1	0.9	0.5	**0.4**
SAMSUM	mean range	**2.0**	2.1	3.2	2.7	6.5
	max range	**3.1**	4.2	5.3	3.3	10.9
	SD	**0.8**	**0.8**	1.2	1.1	2.4
	PAIRGAP	1.0	**0.9**	1.5	1.3	2.8

4.2 H4: NLI is more order-sensitive than sentiment, but the thresholds fail

Table 4 reports sensitivity at $K \geq 4$, where every demonstration set has 8 orders. NLI is more order-sensitive than sentiment for every model: the mean range is 5.9–13.3 points on MNLI and 2.3–6.8 on RTE, against 1.0–4.8 on SST-2. The thresholds, however, fail. No model has a mean MNLI range above 15 points, and RTE never exceeds 15 even in its worst demonstration set (12.5). Only the single worst MNLI set for 0.5B (21.7) and 1.5B (17.5) exceeds 15. The "below 5 points on sentiment" part holds on average, but the worst SST-2 set for 0.5B still reaches 15.5 points. **H4 is partially supported**: as a direction, not as thresholds. Because we cap orders at 8 per set, these ranges are lower bounds on the range over all $K!$ orders.

4.3 Are order effects real?

A best–worst gap on 200 test items mixes order effects with item-level noise. Table 5 separates them. On MNLI, for all sizes, an order that does well on one half of the test items tends to do well on the other half (split-half $\rho = 0.47$–0.72), and the observed range is 1.7–2.8$\times$ the null range. The same holds on SAMSUM for models of 3B and above, most strongly at 14B ($\rho = 0.67$, ratio 2.67). On SST-2 for 3B and above, in contrast, the best and worst orders are indistinguishable from re-shuffled noise (ratio 0.89–1.00, $\rho \approx 0$), and RTE at $\geq$3B is close to this regime (ratio 1.11–1.23). For these cells, about 1–2 points of range appear even though order has no systematic effect, so "choosing the best order" on a dev set would overfit.

Numerical noise floor. Table 6 compares identical-prompt reruns with order changes. **For classification, order effects are real.** Reruns flip 0–3% of predictions while order changes flip 1–18%, and numerical noise accounts for at most 6% of order variance in 10 of 12 cells (maximum 23%,

Table 5: Are order effects systematic? Split-half Spearman ρ of per-order scores across two random halves of the test items, and ratio of the observed best–worst range to a null range in which order labels are shuffled within each test item ($K \geq 3$). Values with ratio ≥ 1.5 (order effects clearly above item-level noise) in **bold**.

	Split-half ρ					Observed / null range				
Task	0.5B	1.5B	3B	7B	14B	0.5B	1.5B	3B	7B	14B
MNLI	0.56	0.72	0.53	0.47	0.58	**2.08**	**2.83**	**1.83**	**1.70**	**2.07**
RTE	0.45	0.47	0.18	0.10	0.23	**1.79**	**1.60**	1.22	1.11	1.23
SST-2	0.38	0.30	−0.04	−0.07	0.07	**1.64**	1.41	1.00	0.89	0.98
SAMSUM	0.26	0.22	0.47	0.38	0.67	1.22	1.39	**1.80**	**1.54**	**2.67**

Table 6: Numerical noise floor: identical prompts re-run ($K = 4$, demonstration sets 0–2, all 8 orders). "Noise share" is the fraction of order-induced squared score differences that identical reruns also produce. For SAMSUM, "flips" are the percentage of summaries whose text differs between runs. The 14B model could not be re-run (section 5.1).

Model	Task(s)	Flips, rerun (%)	Flips, order (%)	Abs. gap, rerun (pts)	PAIRGAP, order (pts)	Noise share
0.5B	MNLI/ RTE/ SST-2	2.9 / 2.4 / 0.3	17.7 / 7.0 / 10.8	0.98 / 0.96 / 0.21	3.89 / 2.21 / 6.49	0.06 / 0.17 / 0.00
1.5B	MNLI/ RTE/ SST-2	1.3 / 0.9 / 0.1	9.1 / 3.8 / 2.2	0.52 / 0.56 / 0.10	4.52 / 1.38 / 1.10	0.02 / 0.23 / 0.03
3B	MNLI/ RTE/ SST-2	0.8 / 0.3 / 0.1	10.4 / 2.3 / 3.4	0.54 / 0.17 / 0.08	3.15 / 0.72 / 0.69	0.04 / 0.07 / 0.06
7B	MNLI/ RTE/ SST-2	0.5 / 0.3 / 0.0	5.8 / 2.9 / 1.0	0.25 / 0.25 / 0.00	1.85 / 1.20 / 0.42	0.03 / 0.06 / 0.00
0.5B / 1.5B / 3B / 7B	SAMSUM	53 / 35 / 38 / 35	–	0.47 / 0.53 / 0.40 / 0.51	0.84 / 0.80 / 2.31 / 0.90	0.30 / 0.42 / **0.03** / 0.35

RTE 1.5B). **For generation, numerical noise is a large confound.** Between 35% and 53% of greedy summaries differ textually between identical runs, and for 0.5B, 1.5B and 7B, 30–42% of the ROUGE-L order variance at $K = 4$ is reproduced by reruns alone. Only 3B (noise share 0.03) shows clearly above-noise order effects at $K = 4$. This weakens the SAMSUM sensitivity numbers in table 4 and table 9 for those sizes. We could not re-run 14B (section 5.1); its high split-half reliability ($\rho = 0.67$) argues that its order effects are systematic, since numerical noise should not be consistent across halves of the test set.

4.4 H2: position influence in classification depends on task and size

Is the last demonstration 2–3× as influential as the first? Table 7 and figure 2 show that this holds in only 2 of 15 model–task cells at $K = 8$: SST-2 with 1.5B (2.42×, CI [1.83, 3.50]) and RTE with 7B (1.81× [1.27, 2.44]; 2.22× [1.71, 2.97] at $K = 4$). For 0.5B on SST-2, influence rises steeply toward the end (0.73 nats at the last position), but the first position is slightly negative, so this is recency in shape without a meaningful ratio. Several cells point the other way: on SST-2, 7B (0.49×) and 14B (0.57×) weight *early* demonstrations more.

Anti-recency on NLI. For 1.5B and 3B on RTE and MNLI, flipping the *last* demonstration's label lowers the probability of the flipped-to label. For 3B on MNLI at $K = 4$, the last position has influence -0.79 nats against $+0.61$ for the first, and at $K = 8$ the flip moves predictions 4.2 percentage points *away* from the flipped label. The effect is concentrated in the final position: influence is flat over earlier positions and then turns negative (figure 2). One possible mechanism is that when the demonstration closest to the test item looks inconsistent, the model contrasts away from it rather than copying it; we did not test this. Unmanipulated prompts agree: for 3B on MNLI at $K = 8$, the prediction matches the label at position 1 for 36.4% of items and at position 8 for 30.8% (chance ≈33%); for 14B, 36.8% vs. 31.8%. Only 0.5B on SST-2 shows clear recency here (47.8% vs. 52.8%).

The 14B model is roughly position-uniform on RTE and MNLI (ratios 1.19 and 0.93, CIs include 1) and mildly primacy-weighted on SST-2. **H2 is not supported** as a general claim: the direction of the position effect depends on both model and task. We also fit an observational regression of the label score on the label at each position (appendix A); its absolute per-position values are poorly identified because they come mostly from label-composition differences between 10–20 demonstration sets, so we treat the causal probe as primary.

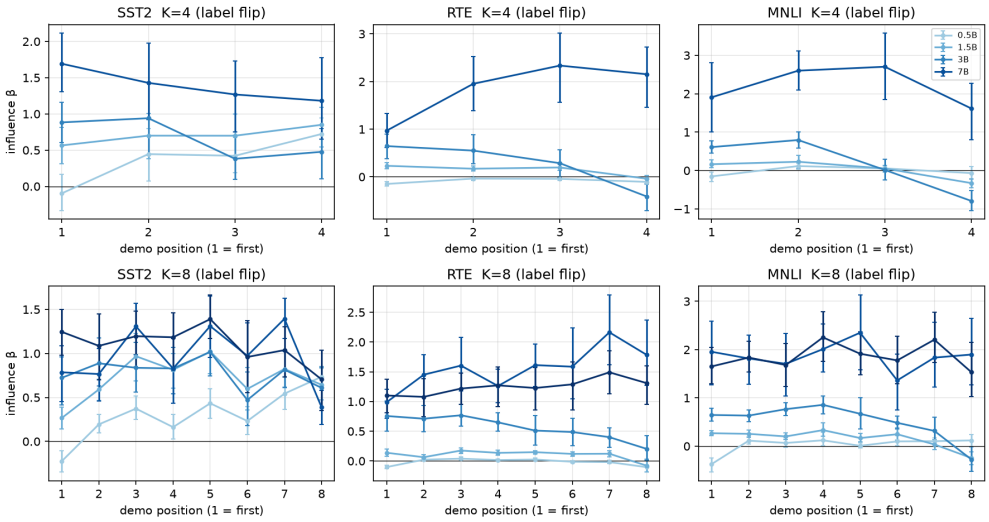


Figure 2: Causal label-flip influence by demonstration position (1 = first) at $K = 4$ (top; 0.5B–7B) and $K = 8$ (bottom; all sizes, darker = larger). Error bars: 95% bootstrap CI. Influence rises toward the end only for small models on SST-2 and for 7B on RTE. On RTE and MNLI, 1.5B and 3B influence turns *negative* at the last position (anti-recency), while 7B and 14B are roughly flat.

Table 7: H2: causal label-flip influence at $K = 8$ (nats; increase in $\log P$ of the flipped-to label, averaged over test items, 10 demonstration sets $\times$ 2 orders). The last/first ratio has a 95% bootstrap CI over demonstration sets. Ratios consistent with H2 (CI above 1 and overlapping 2–3) in **bold**. "n/a": first-position influence is negative or near zero, so the ratio is not meaningful.

Task	Model	Pos. 1	Pos. 2–7 (mean)	Pos. 8	Last/first [95% CI]
SST-2	0.5B	−0.23	0.30	0.73	n/a
	1.5B	0.27	0.80	0.64	**2.42** [1.83, 3.50]
	3B	0.72	0.81	0.61	0.84 [0.52, 1.37]
	7B	0.79	1.10	0.39	0.49 [0.25, 0.87]
	14B	1.25	1.14	0.70	0.57 [0.29, 0.88]
RTE	0.5B	−0.10	0.01	−0.10	n/a
	1.5B	0.14	0.13	−0.08	−0.56 [−1.75, 0.23]
	3B	0.76	0.59	0.20	0.27 [0.02, 0.52]
	7B	0.99	1.61	1.79	**1.81** [1.27, 2.44]
	14B	1.10	1.26	1.31	1.19 [0.88, 1.60]
MNLI	0.5B	−0.37	0.08	0.12	n/a
	1.5B	0.27	0.21	−0.25	−0.92 [−1.46, −0.46]
	3B	0.65	0.62	−0.28	−0.43 [−0.90, −0.03]
	7B	1.95	1.85	1.90	0.97 [0.63, 1.52]
	14B	1.65	1.95	1.53	0.93 [0.58, 1.37]

4.5 H3: generation shows recency, not primacy

Does the first demonstration dominate generation? Table 8 and figure 3 show that it does not. For 0.5B, 1.5B and 3B, the influence of a style marker rises monotonically with position: the first/last ratio is 0.64–0.74 at $K = 6$ and 0.72–0.80 at $K = 4$, with all CIs excluding 1. At 7B the profile is essentially flat, with a statistically detectable but tiny primacy (first/last 1.02 at $K = 6$, 1.03 [1.02, 1.05] at $K = 4$, i.e., about 3%). At 14B it is position-independent (0.98 [0.95, 1.01]). **H3 is refuted.** The only primacy we observe (7B) is roughly ten times smaller than the recency seen in the small models (a 2–3% excess against 20–36%). A secondary, non-causal measure, the lexical copying of each demonstration's summary words into the output, is small (under 1.5% of output words) and noisy, with no consistent primacy. Because the probe aggregates many paired comparisons on log-probabilities rather than on sampled text, it is less affected by the numerical noise of section 4.3.

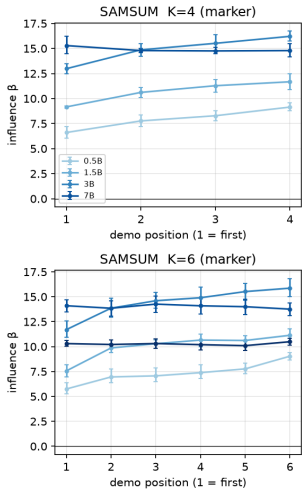


Figure 3: Style-marker influence on SAMSUM by demonstration position at $K = 4$ (top; 0.5B–7B) and $K = 6$ (bottom; all sizes). Error bars: 95% bootstrap CI. For 0.5B–3B, a marker in a later demonstration is copied more often (recency). For 7B and 14B influence is flat across positions. No model shows the strong primacy predicted by H3.

Table 8: H3: style-marker influence on SAMSUM at $K = 6$ (nats; increase in $\log P$ that the output begins with "In short," when demonstration i's summary begins with it, relative to no marker). Largest influence per model in **bold**. H3 predicts a first/last ratio above 1; small models show the opposite.

Model	Pos. 1	2	3	4	5	6	First/last [95% CI]
0.5B	5.75	6.96	7.07	7.39	7.77	**9.04**	0.64 [0.59, 0.68]
1.5B	7.56	9.87	10.29	10.66	10.61	**11.13**	0.68 [0.62, 0.74]
3B	11.70	13.86	14.60	14.90	15.52	**15.86**	0.74 [0.72, 0.75]
7B	14.10	13.85	**14.26**	14.10	14.00	13.76	1.02 [1.00, 1.05]
14B	10.30	10.21	10.31	10.20	10.10	**10.49**	0.98 [0.95, 1.01]

4.6 H5: scale helps on easy classification, not everywhere

Table 9 and figure 4 show that larger models are less order-sensitive on SST-2 and RTE: PAIRGAP falls monotonically from 2.10 to 0.39 points on SST-2 and from 2.96 to 0.92 on RTE (Spearman $\rho = -1$), and flip rates fall from 4.6% to 0.6% and from 9.1% to 1.9%. On MNLI the trend is non-monotone: sensitivity is lowest at 7B (PAIRGAP 2.37) and higher again at 14B (3.60). On SAMSUM the trend points the other way: 14B is the most order-sensitive model (PAIRGAP 2.29 vs. 0.96 for 0.5B), and its order effects are the most reproducible (split-half $\rho = 0.67$, table 5). **H5 is supported for SST-2 and RTE, but not universally.** With only five sizes from one family, the trend tests are coarse: $\rho = \pm 0.7$ is not significant ($p = 0.19$).

5 Discussion

What survives from the hypothesis. Three parts hold: NLI is more order-sensitive than sentiment for every model; larger models are usually less order-sensitive on classification; and order variance is flat at higher K. Three parts do not. The 4-shot threshold disappears once sensitivity is measured with a metric that does not grow with the number of sampled orders; the plateau is reached at $K = 2$. The 15/5-point thresholds fail on average. And the recency (classification) vs. primacy (generation) split is reversed in places: recency appears where it was not predicted (summarization with small models) and fails where it was predicted (NLI with mid-size models).

Relation to prior work. The recency bias that Zhao et al. [2021] documented in GPT-3 base models appears here only in the smallest instruction-tuned models on SST-2; instruction tuning and

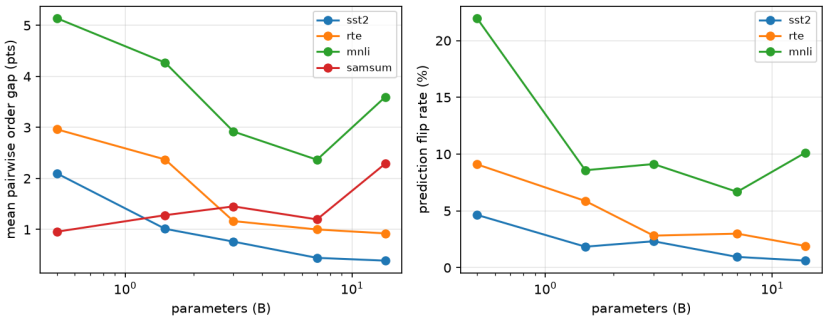


Figure 4: Order sensitivity vs. model size (log scale). *(Left)*Mean pairwise order gap ($K \geq 2$). *(Right)*Prediction flip rate between two orders. Sensitivity falls monotonically with size on SST-2 and RTE, reaches its minimum at 7B on MNLI, and rises with size on SAMSUM.

Table 9: H5: order sensitivity by model size ($K \geq 2$). PAIRGAP in points; flip rate is the percentage of test predictions that change between two orders. Spearman ρ is over the 5 sizes. Lowest per row in **bold**.

Task	Metric	0.5B	1.5B	3B	7B	14B	ρ [p]
SST-2	PAIRGAP	2.10	1.01	0.76	0.44	**0.39**	−1.0 [<.001]
	flip rate (%)	4.6	1.9	2.3	0.9	**0.6**	−0.9 [0.04]
RTE	PAIRGAP	2.96	2.37	1.16	1.00	**0.92**	−1.0 [<.001]
	flip rate (%)	9.1	5.9	2.8	3.0	**1.9**	−0.9 [0.04]
MNLI	PAIRGAP	5.14	4.27	2.92	**2.37**	3.60	−0.7 [0.19]
	flip rate (%)	22.0	8.6	9.1	**6.7**	10.1	−0.3 [0.62]
SAMSUM	PAIRGAP (ROUGE-L)	**0.96**	1.28	1.45	1.20	2.29	+0.7 [0.19]

scale may remove it, although our design cannot separate the two. Lu et al. [2022] reported that order sensitivity does not vanish with scale. We find a more mixed picture within one family: it shrinks substantially on easy tasks, persists on MNLI, and grows for summarization at 14B. The anti-recency effect on NLI is, to our knowledge, not described in prior work and deserves a targeted study.

Practical implications. When is order worth tuning? Our reliability analysis gives a direct answer. For capable models (≥3B) on easy tasks such as SST-2, order-induced differences are about 1 point and are no larger than item-level noise, so tuning order on a dev set would overfit. Where order effects are systematic (MNLI at every size, and summarization at ≥3B), averaging over orders or searching for an order is justified. We also recommend that papers report a pairwise or variance-based sensitivity measure, rather than the best–worst range, when comparing across shot counts or across setups with different numbers of sampled orders, and that they check whether the "best" order is reproducible on held-out items.

Error analysis. The 0.5B model's summaries usually run to the 80-token cap with repetitive text (median 67 words vs. 24 for 14B), which explains its low and flat ROUGE-L. The negative influence on NLI is concentrated in the final position (figure 2), which suggests a mechanism specific to the demonstration adjacent to the test item rather than a gradual decay.

5.1 Limitations

- **One model family and one template per task.** We test only QWEN2.5-INSTRUCT (released in 2024) with plain-text prompts and deterministic decoding. Other families, chat-formatted prompts or frontier API models may behave differently; no API models were available on our host.
- **Small test sets.** With 200 classification items and 80 dialogues, per-order accuracy carries roughly ±3 points of item-level noise. Section 4.3 quantifies this, but larger test sets would sharpen per-cell conclusions.

- **Limited orders and sets.** We cap orders at 8 per set, so best–worst ranges are lower bounds on the range over all $K!$ orders ($8! = 40{,}320$). 7B and 14B use 10 demonstration sets per K rather than 20, and the 14B probes use only $K = 8$ (classification) and $K = 6$ (SAMSUM).
- **Narrow generation probe.** The style-marker probe measures one kind of influence: copying a surface marker. We do not test primacy in content selection or in the ordering of facts, and SAMSUM is our only generation task.
- **Non-deterministic inference.** Batched bf16 inference accounts for about a third of the measured SAMSUM order variance for 0.5B, 1.5B and 7B. The 14B weights (≈30 GB) did not fit in the free memory of the shared GPUs (≤23 GB) during the noise-floor session, so the 14B summarization result lacks a noise-floor check.
- **Coarse size trends and contamination.** Size trends rest on 5 points. SST-2, GLUE and SAMSUM are public and likely seen in pretraining, which affects absolute accuracy more than order effects.

Broader impact. This work concerns evaluation methodology. Its main risk is misuse in the opposite direction from our advice: selecting a favorable demonstration order to inflate reported few-shot results. Our reliability test is intended to make such selection visible.

6 Conclusion

We studied how the order of few-shot demonstrations affects in-context learning across shot counts, task types and model sizes, using one controlled protocol on five QWEN2.5-INSTRUCT models. Demonstration order changes performance by about 1–13 points of mean best–worst range, depending strongly on task and model. Order variance is at its plateau already at two shots; the widely reported rise up to four shots comes from the number of orders sampled. Position influence is task- and size-dependent rather than a universal recency/primacy split: we see recency for small models on sentiment and on summarization, anti-recency for mid-size models on NLI, and roughly uniform influence for the largest models. Scale reduces order sensitivity on simple classification but not on MNLI or summarization. The key takeaway is that whether a "best order" exists is itself an empirical question: for strong models on easy tasks it is mostly noise, while on harder tasks it is systematic and worth handling.

Future work should replicate these measurements on other model families and with chat templates; test whether NLI anti-recency comes from contrast with an inconsistent nearest demonstration, by flipping to correct vs. incorrect labels; use larger test sets and full permutation enumeration for $K \leq 5$ to estimate true worst cases; probe content-level primacy in generation; and test whether ensembling over orders recovers the systematic losses on MNLI.

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A Additional Details and Figures

Example prompt. An RTE prompt with $K = 2$:

```
Decide whether the premise entails the hypothesis. Answer yes or no.

Premise: ...
Hypothesis: ...
Answer: no

Premise: ...
Hypothesis: ...
Answer: yes

Premise: <test>
Hypothesis: <test>
Answer:
```

Reproducibility. We use seeds 42 (design), 1042 (extension sets), 7 (probe label choice) and 0 (bootstrap). Software: vLLM 0.31, PyTorch 2.13, CUDA 13, pandas, SciPy and rouge-score. Classification scoring uses `allowed_token_ids` with `logprobs_mode=processed_logprobs`. Main sweeps on the shared A6000 took about 5 (0.5B), 7 (1.5B), 30 (3B), 74 (7B) and 68 (14B) minutes; probes took 5–50 minutes each.

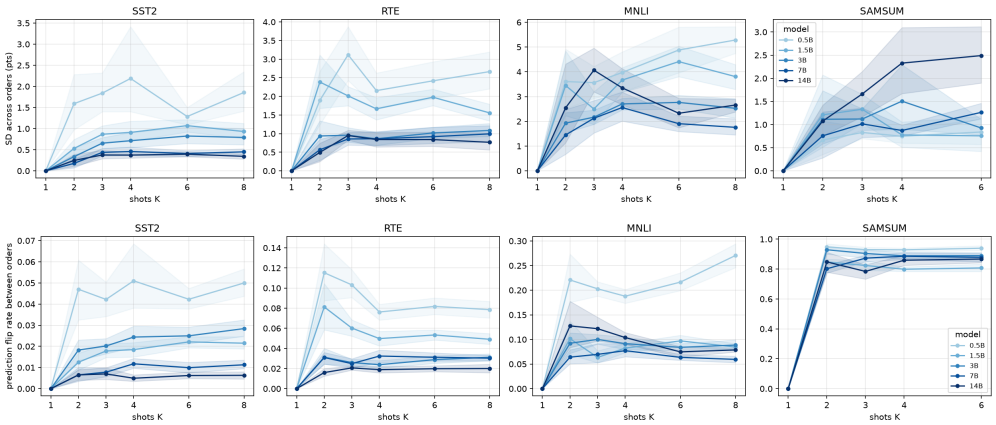


Figure 5: *(Top)*Standard deviation of the metric across orders vs. K. *(Bottom)*Prediction flip rate between two orders vs. K (classification).

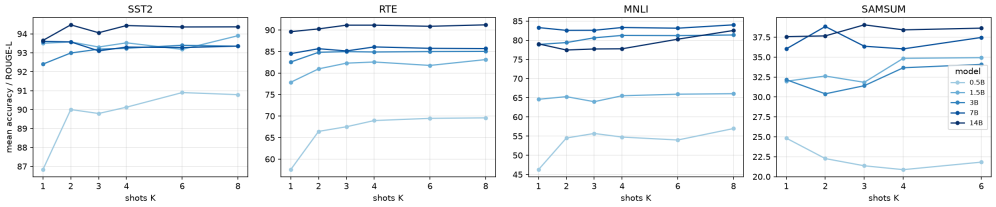


Figure 6: Accuracy (ROUGE-L for SAMSUM) vs. shot count for each model.

Additional sensitivity measures. Figure 5 shows the standard deviation across orders and the prediction flip rate as functions of K. Both behave like PAIRGAP: roughly flat from $K = 2$.

Accuracy by shot count. Figure 6 plots the full accuracy curves summarized in table 2.

Observational position estimates. Figure 7 shows the observational estimates: a regression of the label score on the label at each position (classification), and regressions of output length on demonstration summary length and lexical copying by position (generation). Absolute per-position values in the classification regression are poorly identified because they come mostly from label-composition differences between 10–20 demonstration sets, which are confounded with demonstration content; only within-set last-minus-first contrasts are clean. Lexical copying is under 1.5% of output words and shows no consistent primacy.

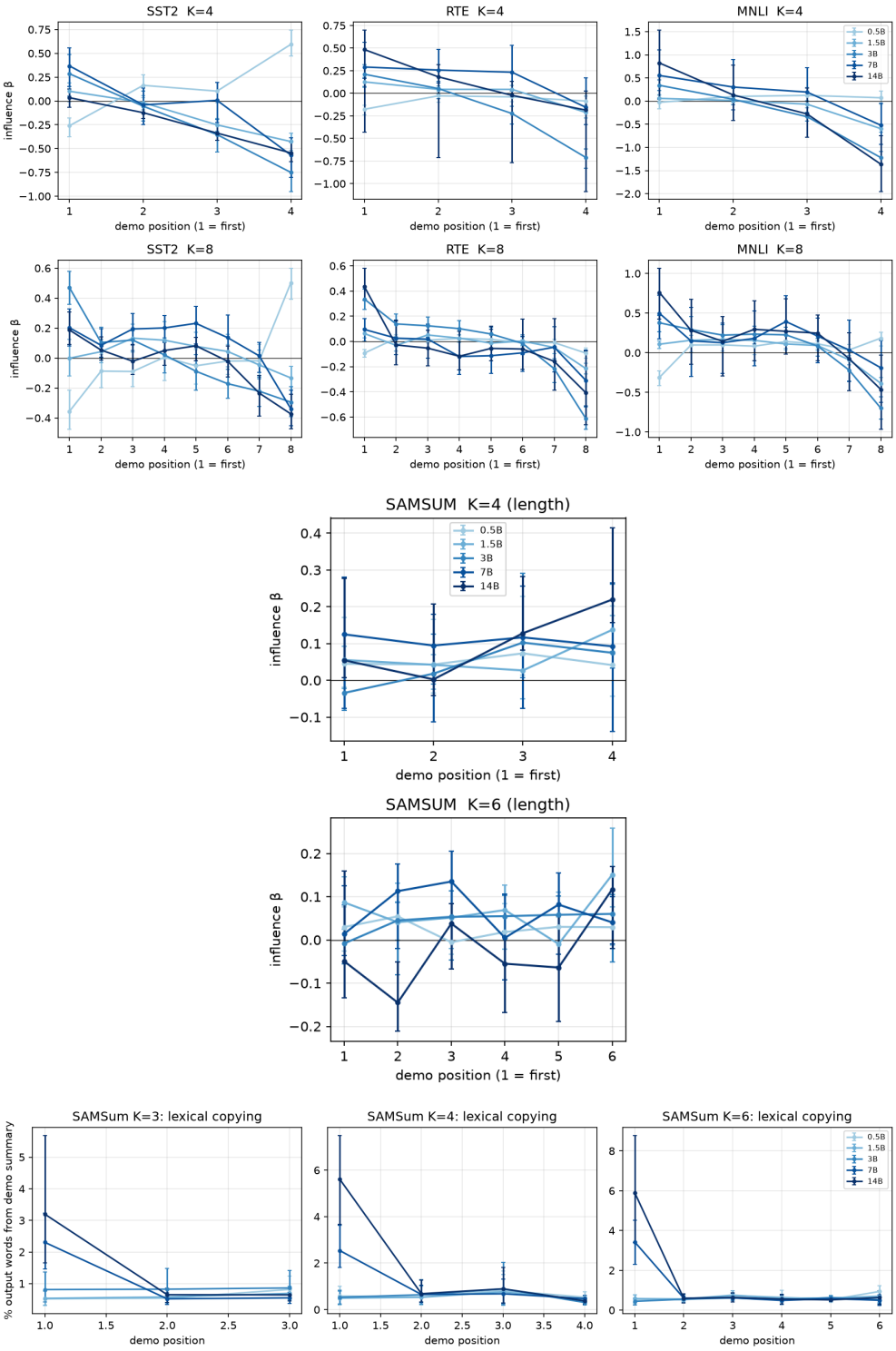


Figure 7: Observational (non-causal) position estimates. *(Top)*Regression-based classification influence. Middle: regression of output length on demonstration summary length by position (SAMSUM). *(Bottom)*Lexical copying of each demonstration's summary words into the output, by position (SAMSUM).